AI-Powered Smart Dustbin for Efficient Waste Classification

Project ID - 199568

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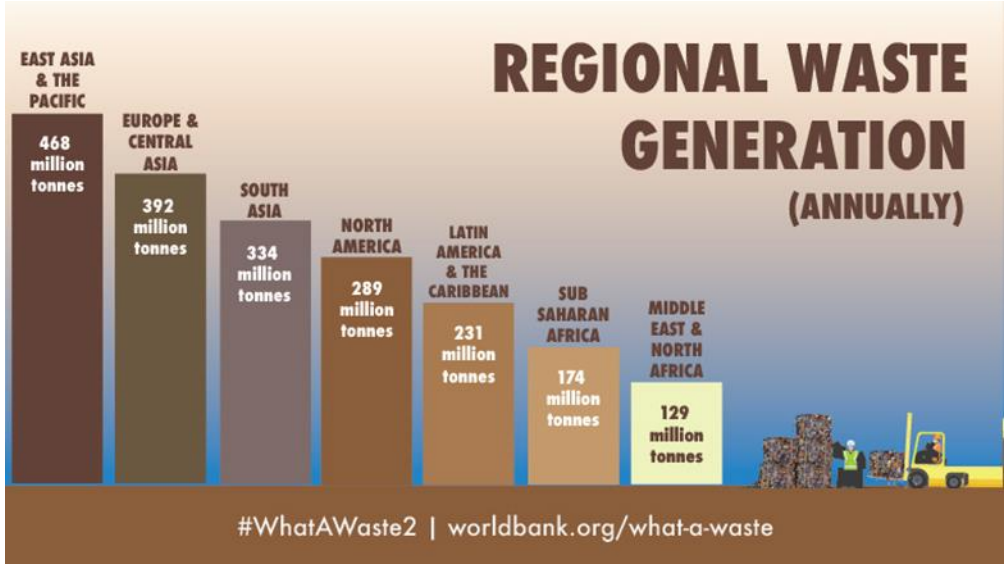
City- Mumbai

State - Maharashtra

Introduction

- Over 2 billion tons of waste generated globally each year.
- Improper disposal leads to pollution in air, soil, and water, contributing to climate change.
- Health risks arise from exposure to hazardous waste, especially in low-income areas.
- Valuable resources in waste, like metals and plastics, often go unrecovered, depleting natural resources.
- Manual waste sorting is labor-intensive, time-consuming, and prone to human error.
- Increasing complexity of waste types (e-waste, organic, plastic, metal) complicates sorting and recycling.





Objectives

01

Develop an AI-Powered Smart Dustbin: Enable autonomous waste classification and sorting to boost efficiency and sustainability.

02

Automate Waste Classification: Use AI to identify and categorize dry, wet, and electronic waste, improving sorting accuracy.

03

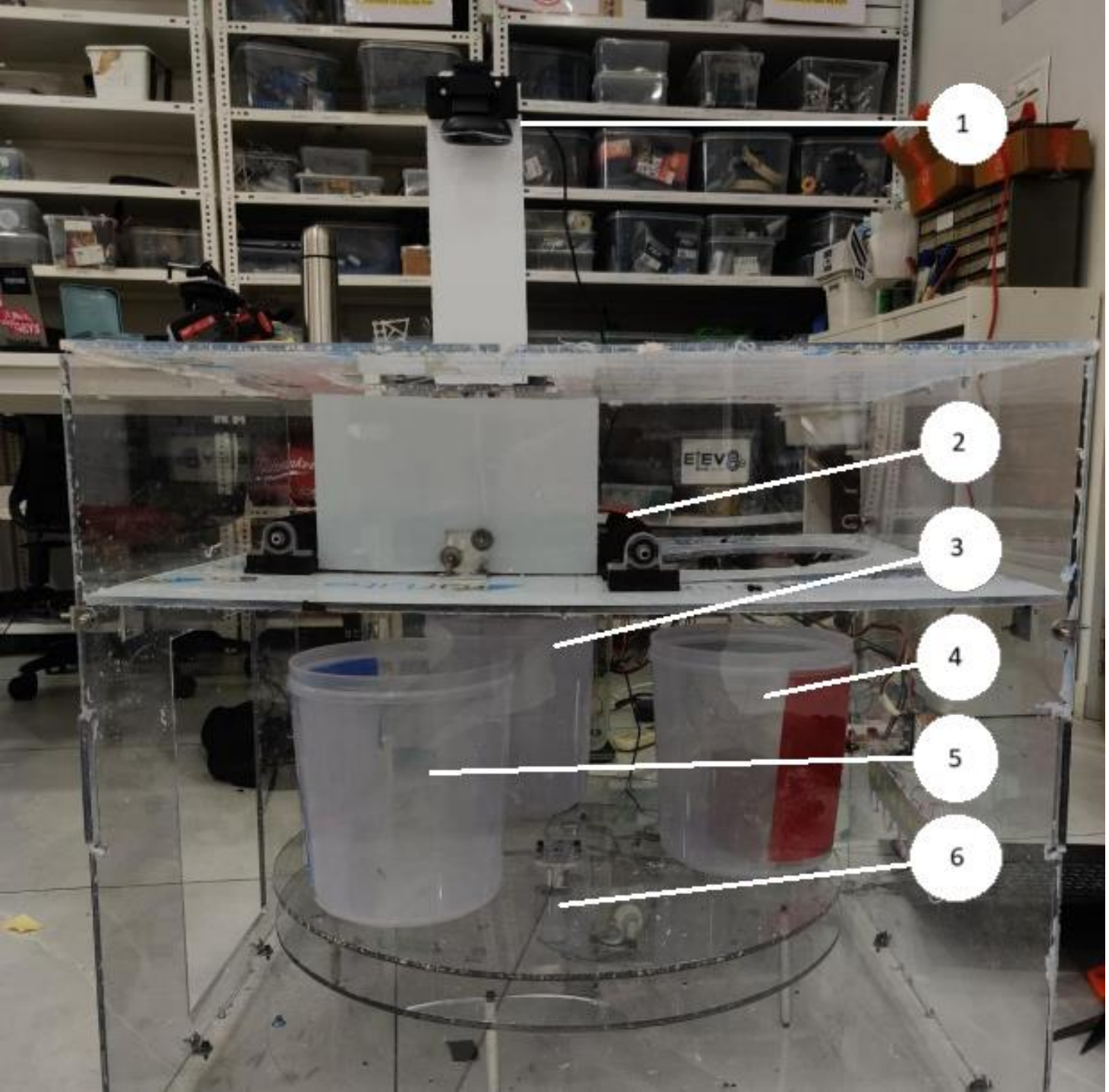
Increase Recycling Rates: Enhance sorting precision to ensure more recyclable materials are correctly processed.

04

Reduce Environmental Impact: Promote correct disposal to minimize landfill waste and pollution.

05

Streamline Waste Disposal: Use an automated system to direct waste to bins, reducing manual labor.

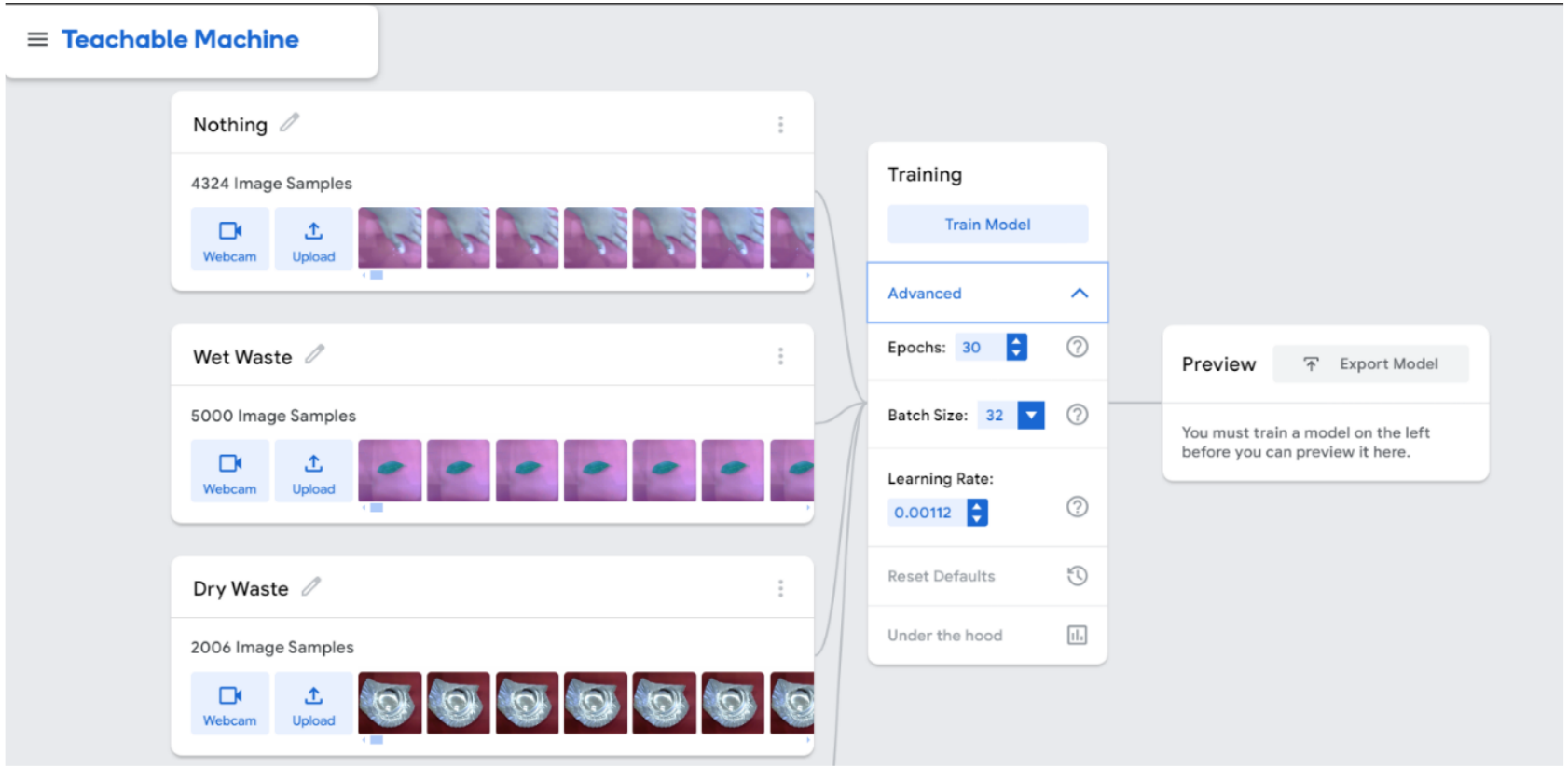


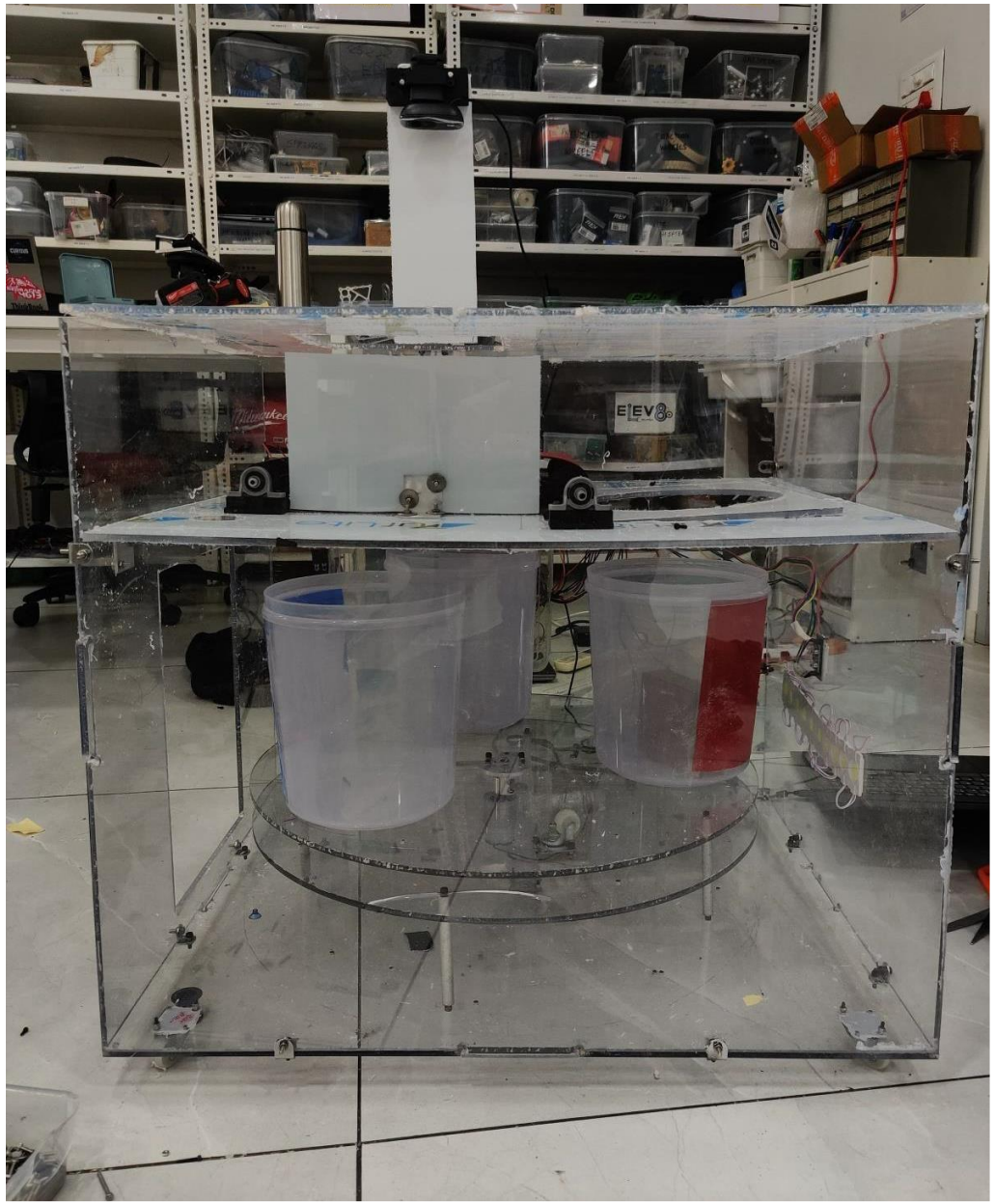
Hardware
Components for Smart Dustbin:

1. Camera
2. Conveyor belt
3. Wet waste bin
4. Dry waste bin
5. Electronic waste bin
6. Rotating disc motor

Google Teachable Machine

- User-friendly tool to easily capture images of various waste types and create a comprehensive dataset.
- Using this dataset, I trained the model to recognize and classify waste accurately, providing the foundational intelligence for the system.





Acrylic Structure

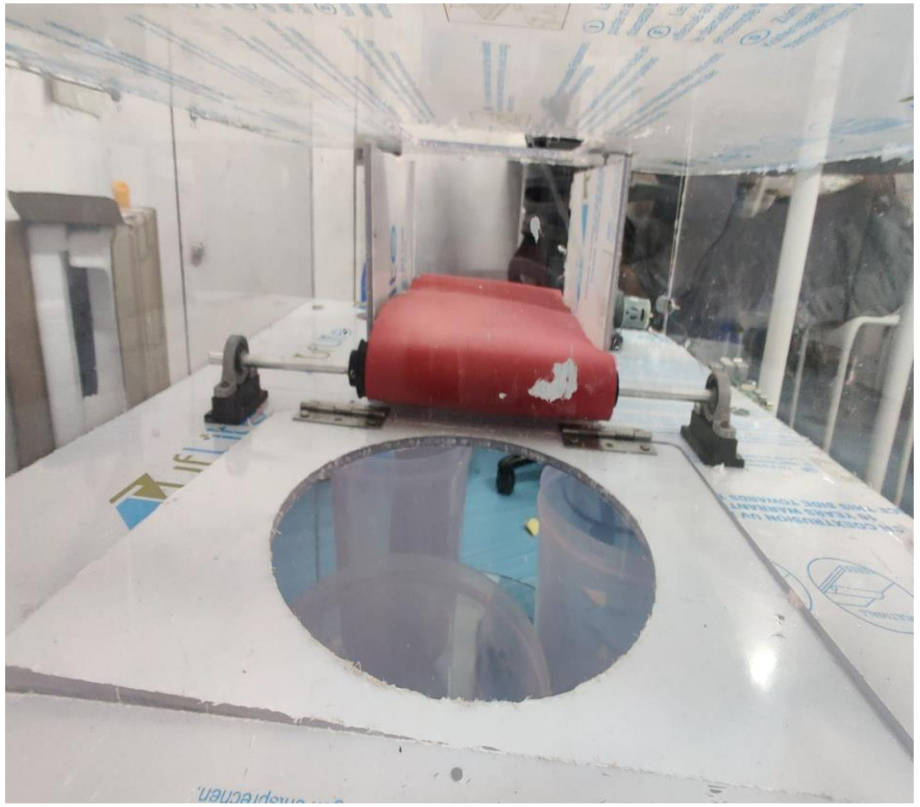
- The entire device is constructed from **transparent acrylic**, allowing for visibility of the internal mechanisms.
- This design choice not only adds an aesthetic element but also facilitates understanding of the operational process.
- The acrylic components were shaped using CNC machining, ensuring precise dimensions and a clean finish.

Conveyor Belt

- The **conveyor belt** serves as the primary transport mechanism for moving waste towards the sorting area.

- This continuous movement allows for a steady flow of items, ensuring that each piece of waste is captured and classified efficiently.

- The conveyor belt is driven by a DC motor, which is controlled through the Cytron MDD 10A Motor Driver, enabling precise speed regulation

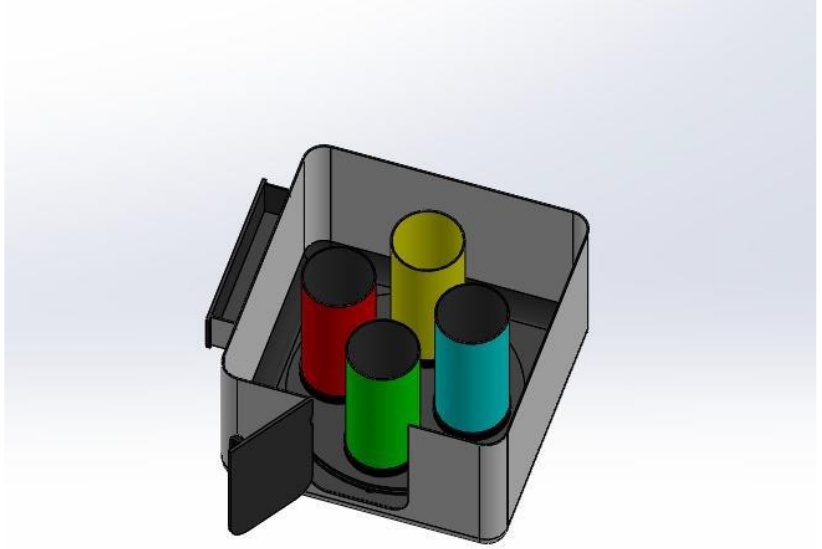


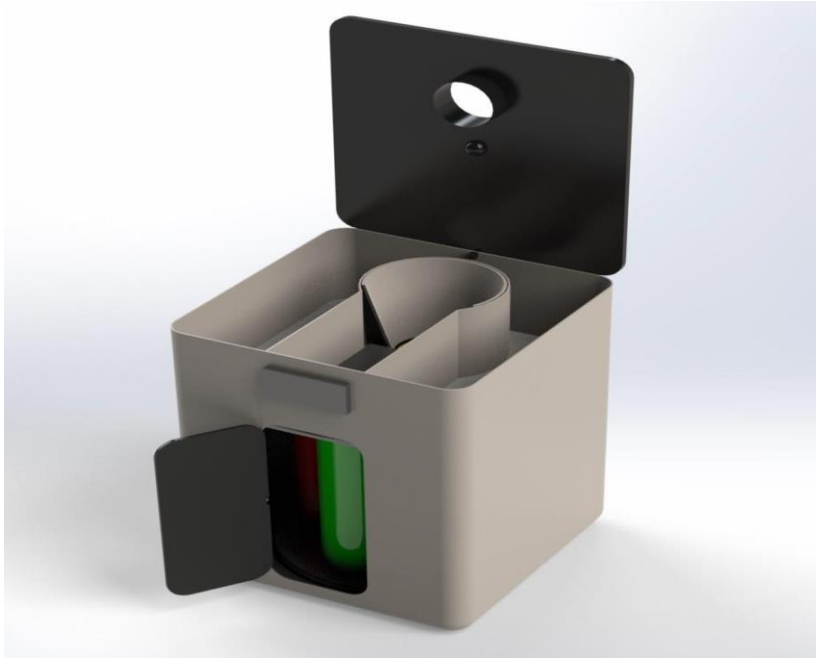
Rotating Disk with Three Bins

- Below the conveyor belt is a **rotating disk** equipped with three designated bins for sorting waste. Once the type of waste is classified, the disk rotates to align the appropriate bin underneath the conveyor.

- This mechanism is crucial for directing waste to its correct disposal location, whether it be for dry, wet, or electronic materials.

- The integration of this rotating disk enhances the system's overall efficiency and effectiveness in waste management.





Code Explanation

- **Library Imports**: Imports NumPy, OpenCV, TensorFlow Lite, and GPIO libraries for numerical operations, image processing, model inference, and hardware control.

```
import numpy as np
from PIL import Image
import cv2
from tflite_runtime.interpreter import Interpreter
import RPi.GPIO as GPIO
import time
```

- **Model Loading**: Loads a TensorFlow Lite model for waste classification (dry, wet, electronic) and prepares it for real-time inference.

```
interpreter = Interpreter(model_path='/home/pi/Desktop/model_unquant.tflite')
interpreter.allocate_tensors()
```

- **Label Loading**: Loads waste category labels from a text file, allowing the system to recognize and respond to each classification.

- **GPIO Setup**: Initializes GPIO pins on the Raspberry Pi for motor control and limit switch readings to manage conveyor and sorting operations.

```
with open('/home/pi/Desktop/labels.txt', 'r') as file:
    labels = [line.strip() for line in file]
labels.append('N')
```

- **Video Capture Initialization**: Starts a video stream from the camera to capture images of incoming waste in real-time.

- **Image Preprocessing and Inference**: Processes captured images to match the model's requirements, including resizing and normalizing, before sending to the model for classification.
- **Label Detection and Classification**: Analyzes model output probabilities to determine waste type; uses a detection counter to confirm consistent classification before sorting.
- **Motor Activation for Sorting**: Activates motors to align and position waste with the correct bin based on classification, with feedback from limit switches for accuracy.
- **System Reset**: Resets motor positions and detection counters after each sorting action to prepare for the next waste item.
- **Startup Automation**: Configures the code to run automatically on startup using rc.local, enabling hands-free operation of the system.

```
GPIO.setmode(GPIO.BCM)

# Set up pins for the limit switches with internal pull-up resistors
LIMIT_SWITCH_PIN_1 = 16
LIMIT_SWITCH_PIN_2 = 5
LIMIT_SWITCH_PIN_3 = 6

GPIO.setup(LIMIT_SWITCH_PIN_1, GPIO.IN, pull_up_down=GPIO.PUD_UP)
GPIO.setup(LIMIT_SWITCH_PIN_2, GPIO.IN, pull_up_down=GPIO.PUD_UP)
GPIO.setup(LIMIT_SWITCH_PIN_3, GPIO.IN, pull_up_down=GPIO.PUD_UP)

# Define GPIO pins for the motor driver
PWM_PIN = 17
DIR_PIN = 27
PWM_PIN2 = 23
DIR_PIN2 = 24

# Set up the GPIO pins for motor control
GPIO.setup(PWM_PIN, GPIO.OUT)
GPIO.setup(DIR_PIN, GPIO.OUT)
GPIO.setup(PWM_PIN2, GPIO.OUT)
GPIO.setup(DIR_PIN2, GPIO.OUT)
```

Conclusions

- The Smart Waste Management System effectively automates waste sorting with high accuracy, reducing the need for manual labor and minimizing errors.
- AI and machine learning integration enable precise classification of waste types, enhancing recycling efficiency and sustainability.
- The automated mechanical system ensures accurate waste placement, streamlining the disposal process for optimal resource recovery.
- This project sets a foundation for future advancements in sustainable waste management, highlighting the role of AI in addressing environmental challenges.

